CHAPTER 7

SCIENCE

AI IS POISED to redefine the possibilities of every creative enterprise and to search for new conclusions in every scientific field. In turn, subsequent explorations are almost certain to refine and enlarge the circumference of human understanding. As discussed in Chapter 1, it is conceivable, perhaps even likely, that AI will push outward in all directions at once, and that its successes on all fronts will be validated, absorbed, and compounded into massive bulkheads of new human knowledge.

If the recent past has been defined by human triumphs in the engineering of complicated systems—microprocessors, the internet, jet engines, particle accelerators—the future will be defined by AI's engineering of high-dimensional, complex systems: human economies, biological life, and the climate of entire planets, others' and our own.[1]

THE GARDEN OF MEDICINE

The fragility of human health has been responsible for more premature deaths and unnecessary pain than any conflict or natural disaster in history.[2] Though over the past two centuries we have made crude and incomplete attempts to decipher and control the code of life, our ability to proceed with greater precision and intent has been impeded by a single missing ingredient: namely, an

intelligence capable of understanding that code at a sufficient level of detail. Now, propelled by the emergence of an intelligence that far exceeds our own, we are converging upon a revolution in biology that may change our conception of human life.

Founded in the seventeenth century by an edict of King Louis XIII of France, and placed under the authority of his royal physician, *Le Jardin royal des plantes médicinales*—the Royal Garden of Medicinal Plants—was the largest and most advanced such project of its time. Scientific expeditions to places as far-flung as Java and the Amazon had returned with a dazzling variety of plants, meticulously studied for their potential medicinal use by a dedicated staff of botanists.

Less than a decade later, the French monarch himself would succumb to tuberculosis; no earthly antidote had been strong enough to ease his pain or delay his death. But the garden did successfully produce treatments and cures found nowhere else in the world.[3] Today, AI has the potential to be a similarly supernal library of pharmaceuticals, opening for human benefit a vast new repository of remedies for the alleviation of sickness and strain.

Extraordinarily adept at generating new combinations of defined items and identifying their highest-performing attributes, revolutionary AIs like AlphaFold from DeepMind—with its gigantic database of over 200 million protein structure predictions—have opened new vistas in global health. AI is likely to produce breakthroughs not only in the engineering of additional proteins, including new hormones, enzymes, and antibodies, but also in identifying the molecular causes of various diseases and developing potential treatments for them. In turn, medical care, relying on AI's unprecedented resolution at the molecular and genomic scale, could become increasingly personal—with drugs, along with their methods of delivery, tailored to each individual's unique metabolic profile, risk of addiction, estimated tolerances, and susceptibility to potential side effects.

Human doctors committed to the relief of human suffering would thus have a partner to aid them in fulfilling their merciful calling. Machine instruction could guide the most skilled of human hands—as

in neurosurgery—to perform operations long thought possible but too risky to be carried out safely. AI is already helping to find nondestructive or nonintrusive paths through the brain to remove a source of debilitation mechanically or to cure it biologically. Where problems are not physical but psychological in nature, AI could combat cognitive loss, mental illness, psychiatric disorders, and possibly even loneliness.

Indeed, and much to be hoped, AI could also move us away from treatment and closer to *prevention*, thus reducing the need to cure at all. Acting as vigilant early-warning mechanisms, AI systems could alert us to malignancies and abnormalities long before they evolve into serious threats. At the societal level, too, AIs might become advanced health-monitoring systems, capable of identifying and neutralizing infectious diseases before they evolve into planetary pandemics.

Yet all the above scenarios, even the one related to preventing disease and premature death, are examples of remediation. They are ways in which AI might help us to mitigate or solve issues that currently drag down individual health below existing standards of adequate well-being. But what of advancements that could redefine the *maximum* of human health?

Magnified by AI, some medical advances will move from therapies to extensions of human longevity. The recent phenomenon of gene editing demonstrates the nearness and feasibility of such breakthroughs. Scientists wielding the biotechnology known as CRISPR-Cas9—and another variation called prime editing—begin by identifying a specific genetic sequence that they would like to manipulate. A predesigned strand of RNA can then guide another special enzyme to a targeted piece of DNA on that sequence, opening it to make the necessary changes and corrections. Using such tools, and their successors, it may become possible to conquer not only our deficiencies but mortality itself.

Death, however, has always been humanity's divinely imposed limit, with no attempt at exemption left unpunished. Consider the myth of Sisyphus, the crafty and cunning king of ancient Ephyra (now Corinth), who was to be imprisoned in the underworld for

angering the gods. He escaped mortality by a trick—binding Thanatos, the god of death himself, in his own shackles—before fleeing back to the land of the living. But without death to take anyone (or anything) away, all hell broke out on Earth. The old and sick suffered with no end, cattle could not be slaughtered for consumption, and animals could not be killed for sacrifice to the gods.

There's more to the tale, but the crisis eventually ended when Ares, the Greek god of war, intervened, freeing Thanatos and enabling Sisyphus to escape from the deep abyss of Tartarus a second time. But that was it: A cautionary example was made of the Greek king before a third transgression could succeed.

Death remains the great equalizer of life. Even the Chinese emperor Qin Shi Huang[4]—a rumored son of heaven and an early seeker of the elixir of life—died from drinking too much mercury and was buried among his vast army of clay. Seeking immortality, it sometimes seems, only serves to hasten its opposite.

Besides, our impermanence has its benefits. It focuses the mind and lends greater urgency to our endeavors. As the American writer Jack London declares triumphantly: "The proper function of man is to live, not to exist. I shall not waste my days in trying to prolong them. I shall use my time."[5] For others, writes Tolstoy (quoting Socrates), "We move closer to the truth only to the extent that we move further from life."[6] The great physicist John von Neumann, when dying of terminal cancer, requested that a Catholic priest accompany him in his final days—his faith no longer at odds with the agnostic science he had pioneered.[7] Are there not uses for endings?

Today or tomorrow, we might find ourselves required to *check* the degree to which AI can extend our lives. A reduction in our awareness of impending mortality could incidentally effectuate untold changes in the human psyche. In due time, societies may need collectively to decide the ideal length of a human life, and in so doing to answer the attendant metaphysical and spiritual questions: Is human longevity merely the product of a communal expectation, a limit that may be cast aside as a misguided, self-imposed boundary

on our species' potential? Or should we consider the human lifespan, whether naturally or divinely given, a sacred restraint on the power of any single person? These questions run deeper than any individual quest for optimal biology.

Even if the total length of life were not altered, in the future we could perhaps guarantee that life would not end by way of a premature decline born of biological vulnerability. But going too far toward the elimination of that weakness might also have side effects. Was it not our triumphing over challenges, including illnesses, that made us respectable? On the other hand, even if we cure all human diseases or engineer ourselves to be impervious to them, humans will still be vulnerable in other ways: to a physical accident, to financial ruin, to heartbreak.

On a visit to Paris in 1833, Ralph Waldo Emerson was transfixed by the Royal Garden of Medicinal Plants, stunned by how much more advanced the natural world appeared—in all its variety and form—than the machines of the Industrial Revolution.[8] Disoriented in a technologically chaotic age, the transatlantic mood was an ambiguous mix of awe and alarm. Emerson, upon return from that "celebrated repository of natural curiosities," retreated to rural Massachusetts and, inspired by what he had seen, articulated a human response that put the natural world back at the center of the now-mechanized one. His diagnosis emphasized how biology could guide us through unfamiliar times while still serving as our strongest reminder that humans remain the ultimate "definer and map-maker of the latitudes and longitudes of our condition."[9]

A half-century after Emerson's visit, and two and a half centuries after its founding, through periods of violent revolution and curatorial revision, the garden was transformed beyond the immediate needs of its first patron to become a museum for the history of what was then a controversial new idea. Its name: *La Galerie de l'Évolution*, the Gallery of Evolution.[10]

Evolution is set to be redefined in the age of AI, as certain tools raise the prospect of human self-engineering. For instance, most of the gene editing made possible by modern instruments is limited to

somatic or nonproductive cells. Some editing, however, can be done to germ line cells, the characteristics of which are inherited reproductively. Some people conceivably could decide to "correct" congenital diseases in their offspring. Others could go further, choosing to install congenital advantages—advantages that may not belong to either biological parent or, in the extreme, to any other human. That would go beyond the elevation of the human race; it would be the very redesigning of it.

We may soon have the power to determine the pace and direction of our own species. This idea is as controversial today as evolution was in Emerson's time, and the prospect raises the obvious but insidious question: What does the perfect human look like? That question has been asked and answered by various societies, and in some cases has been the basis for "scientific" and political endeavors that brought great human tragedy. So we must also ask, with some trepidation: Should we attempt to find out?

Perhaps such experiments are sacrilegious. Or maybe the human ability to invent these technologies is itself a hint that what we have perceived as our limit was always meant to be broken. If there is a Creator, were we created that we might ultimately create ourselves? If so, is it our duty to ensure our maintenance of human agency in the age of AI? Different communities will come up with different answers to such questions. None will be able to escape the urgent necessity of a response.

ENGINEERING EARTH

The history of Earth is as violent as it is misunderstood. There exists a razor-thin window of climatic conditions suitable for the flourishing of life. Any cooler—as has already happened in no fewer than five ice ages—and our Earth becomes a desolate, frozen rock.[11] Any warmer—as is happening now—and it becomes an infernal hellscape. In Fyodor Dostoevsky's *The Brothers Karamazov*, the devil tells Ivan:

> Our present earth may have been repeated a billion times. Why, it's become extinct, been frozen; cracked, broken to bits, disintegrated into its elements, again "the water above the firmament" [Genesis 1:7], then again a comet, again a sun, again from the sun it becomes earth—and the same sequence may have been repeated endlessly and exactly the same to every detail, most unseemly and insufferably tedious.[12]

While we may not share the fallen angel's exasperation, we are aware of the cyclicality of our planet's geological history. The diverse causes of our planet's five previous mass extinctions—from instantaneous meteor strikes to gradual glacial formation—have all resulted from the extreme vicissitudes of Earth's climate.[13]

Today's problem, of course, is a decisive acceleration toward a new heat extreme. That problem is in fact two separate problems, both deriving from an overreliance on carbon. We authors are convinced (perhaps optimistically) that both are caused by, and ultimately solvable with, chemistry—if its full capacities can be unlocked using AI.

The first problem is an atmospheric warming problem. Our current woes have been brought on by moving too much fossil carbon, too quickly, from the geosphere deep underground to the active biosphere aboveground.[14] Efforts at climate-systems engineering—of which there are two main applications—have always been difficult in theory and even more difficult to test in practice. But where we are hindered, AI may not be. One application is carbon removal, which reverses the transferring of excess carbon—in this case, from our atmosphere back underground into the geosphere. The leading solution here is based on simple chemistry discovered more than two decades ago. AI might identify a new, more efficient method.[15]

Another application is solar geoengineering: the release of certain particles into the atmosphere to reflect sunlight and "cool the Earth." Solar engineering, like carbon removal, could reduce the effects of

climate change, enabling us to avoid some of the most extreme consequences of accumulated carbon. (Neither, however, would address the root causes of climate change.) Other hypothetical ideas for engineering the heavens rely on high-altitude jets and elements erupted by supervolcanoes. If they are ever tried at scale, they, like carbon removal, might turn out to be crude and risky strategies.

An AI, however, if designed to integrate data from instruments on land, in sea, and from space, could create an immensely detailed, real-time model of Earth's climate. At a high level of granularity, our planet's atmospheric chemistry might appear not as the chaotic system we see but rather as just another industrial process of exact chemical inputs and outputs, all capable of precise management.

AI could also respond to one-off incidents that threaten the Earth's precarious climatic balance, whether the eruption of a supervolcano blasting matter into the upper atmosphere or the detonation of atomic bombs that threaten a nuclear winter—though in the latter case it might only prevent ecological collapse while doing nothing about the immediate human catastrophe. Still, by intervening on a planetary scale, an AI accustomed to the precise oversight of our climate would be well-positioned to keep our home habitable.

We therefore believe that AI offers a source of hope in a fight that is too often portrayed as one that we are helpless to win. However, even if AIs prove capable of these interventions, it would be dangerous to become too reliant on them. Such assistance should only ever be considered a supplement and not a substitute.

Moreover, the second problem—the energy problem, as distinct from the atmospheric warming problem—still demands a separate solution. Because hydrocarbons, our predominant source of energy, take millions of years to make but only hundreds of years to consume, humans are now forced to find an alternative for planetary power, irrespective of any changes in the atmospheric composition.[16]

If we are right in general that AI can tune the chemistry of our atmosphere, it may be able to do the same for our energy products,

designing and producing new carbon-free commodities to replace their problematic predecessors. One could imagine an AI testing and tweaking millions of synthetic substitutes in virtual simulation or in a physical lab until arriving at zero-carbon replacements for oil, gas, and coal.

Ideally, AI could also engineer microorganisms optimized to produce these new fuels—and it might design their processes to fit within existing refineries and operate on legacy equipment. In generating energy in the same locations, using the same machinery, but not by the same methods, these sustainable substitutes could be compatible not only with the infrastructures of their production but also with their transportation and consumption. Still another possibility is that AI will finally unlock the viability of fusion; this would solve not only our planet's energy problem but also the energy problem of any other planet on which we might someday find ourselves.

Some will push back on these ideas, finding them unrealistic (especially given the urgency of our immediate challenge) or perhaps undesirable—that is, at odds with our desire to leave nature unchanged, akin to the instinct that so strongly motivates our efforts to preserve it. As the Indigenous Ecuadorean environmentalist Nemonte Nenquimo cautions, "The Earth does not expect you to save her, she expects you to respect her."[17]

We believe our optimism comports with the demands of both respect and humility. We also admit that the only thing we know is that the future is unknowable. By no means do we suggest or imply that superintelligence is required to pull us out of the predicament we have created for ourselves. Even absent assistance from machine intelligence, humans are plenty capable of doing so, and we should not halt or slow our ongoing efforts. To the contrary, we should accelerate them. This is especially so as we recognize the tremendous amount of energy that likely will be consumed by the training and inference of AIs—procedures that, if unaccompanied by any of the intentional and directed efforts we have described here, will only exacerbate our already dire circumstances.

At the same time, if silicon can indeed provide additional solutions to problems created by carbon, we would be wise to explore that opportunity. This is particularly so given the enormous potential benefits to the developing world, which suffers most from the problems caused by climate change—and which would also face particular hardship if the world were to impose hard limitations on global consumption of energy.

Too often, AI is regarded as another product of the same destructive philosophy that is to blame for our current climate dilemma. We are concerned that such a view is too shortsighted, and that it could cause us to miss a significant chance to mend our present without stepping backward to a preindustrial past. It is our hope, as in 1940 it was Winston Churchill's, to bear witness as "the New World, with all its power and might, steps forth to the rescue and the liberation of the old."[18]

BEYOND OUR PLANET

Buried deep in the mountains of southwest China, humanity is holding its breath, listening to its own heartbeat, waiting patiently for the first cries to emerge from the sea of cosmic silence. This remote place is home to the largest radio telescope on Earth—a gigantic, smooth metallic dish as large as the peaks hiding it from view—a machine nicknamed the "Eye of Heaven" and custom-built to hunt for life beyond our planet. But although the sheer vastness and antiquity of the universe mean that it should be a cacophony of signals from other civilizations—or at least from the remnants of earlier ones—instead it is almost dead quiet.

We think we know what to look for. We try to guess the technologies that a sufficiently advanced civilization might have developed and to predict which of their signals we could plausibly see or hear. Scanning for flashes of atomic power in the atmospheres of distant worlds, or screening dense star fields for abnormally large physical structures, we are constrained by the

limits of our instruments—including our imagination.

If life exists beyond our own, Chinese astronomers at this outpost might be the first to hear it.[19] But we now wonder if we humans are really all that likely to make first contact. Perhaps instead it will be done by another alien intelligence—one born of our own planet.

Already, AI is helping us to listen and look for extraterrestrial life, sifting through billions of older technosignatures, separating those caused by human interference from potentially foreign sources.[20] Where humans would only hear static, AI might recognize previously indecipherable or overlooked communications.

Galileo once described our universe as a "grand book" written in the language of mathematics. If alien life forms have learned this universal and precise method of representation and reasoning, as we have done, or if their languages, like ours, can be translated into mathematical form, we may be able to interpret—and respond to—their signals. More than likely, this endeavor would at least be assisted, if not led, by AI.

AIs could become much more than passive watchers and receivers—not just as translators, but also as adventurers, beacons, and scouts. AIs could serve as astronauts, going farther than humans could have imagined. Future humans may even accompany their AIs beyond the solar system. Together they might illuminate the existence of long-gone civilizations and uncover the reasons for their extinction—perhaps educating us about potential dangers ahead. AIs familiar with deep space could discover new, abundant organic materials. In the distant future, they might assist in the engineering of planetary megastructures to protect Earth from impact by comets and asteroids or engulfment by black holes. AI systems could engineer planets' atmospheres to accommodate us or assist humans in adjusting their physiology for easier acclimation to another system.

Of course, AI may become the very reason our civilization fails—ending humanity by bringing us into contact with another alien but hostile intelligence. For instance, AI could make possible louder and longer-lasting signals into space, indicating the presence of human

civilization to anyone listening in the dark. It also could scout out other lived-in worlds, greatly increasing the probability of finding one—with no telling the result. Some argue that, if we are to enter the arena of space exploration at all, we should do it only after achieving a level of technological sophistication that will allow us to defend ourselves against any imaginable foe—suggesting that AI is a prerequisite to, and a safeguard after, any encounter with extraterrestrial life. But if we encounter a post-biological intelligent species, our silicon partners would provide no guarantees of a welcome reception.

The other way that AI could spell the end of human civilization, of course, is by becoming misaligned with humanity. (That possibility would seem all the more pressing if we discover evidence of a machine-provoked extinction of biological life on another planet.) For eons, our fascination with life beyond our own planet consumed our writings, animated our religions, and preoccupied our astronomers. We are confident that humanity has thought longer and harder about faraway worlds than about this more likely possibility—a "first contact," on Earth, with creatures of our own creation. We should be just as concerned, if not more so, with understanding and shaping the nature and intention of those here with us as of those far away in space and time—and this will be the subject of extended discussion in the next chapter.

Even if teams of humans and AIs do not discover alien life forms or pioneer a path to alien planets, they may acquire useful and previously alien knowledge. Together, we could attain elevated understandings of the scope of the universe, the nature of space-time, the stability of star systems, and the nature of gravity in astrophysics. AI's mechanical modeling could shrink stellar distances, compress and expand human time on Earth, and warp and stretch our perception of all edges and aspects of our universe.

An AI-enabled study of astrophysics could enable a more profound human understanding of where we came from in unthinkably ancient time. Religious cosmology has long sought to provide philosophical or theological answers to these questions of human origins. What we find in our next phase of universal

exploration—which will necessarily implicate our deepest past—could color our perception of some of humanity's most sacred beliefs. Perhaps there is a reason the holiest Islamic relic is the Black Stone, a meteorite inside the House of God at Mecca, the center of ritual in the Muslim world.[21] Perhaps there is more to the kabbalistic concept of *kefitzat haderekh*, Hebrew for a miraculous shortcut (literally, "a road-leap") between two distant places in a brief time.[22]

In *The Grand Design*, Stephen Hawking and coauthor Leonard Mlodinow write, "The (unobserved) past, like the future, is indefinite and exists only as a spectrum of possibilities."[23] For now, this is the human perception. It remains to be seen whether such precise mechanical intelligences as those we are developing now could unsettle this uncertainty, pinpointing just one reality. Our creations may thus relay a single universal story, one that feels at once ordinary and special, impossibly small, and miraculously divine.